

Inputs in MPC Bi-weekly

MPC Chatbot

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Contribution in Bi-weekly:

Frontend, Model Building & Model Training

Models we tried:(17th March-22nd March)

1. TF-IDF (Term Frequency and Inverse document frequency)
2. Cosine Matrix
3. BERT-based sentence transformer
 - **sentence-transformers/all-mpnet-base-v2**(higher accuracy, larger size)
 - **sentence-transformers/paraphrase-MiniLM-L6-v2** (optimized for paraphrase tasks)
 - **BAAI/bge-large-en-v1.5** (better for retrieval tasks)

Final Model:(25th March-26th March)

Gen-AI: We've used Gemini API for Retrieving the data from the dataset.

Why Gemini:

- Gemini is free
- Easy to implement
- Don't store the data

Problems we faced:

- Chatbot was in Django and the whole project was in React.js and Node.js.
- So, it was difficult for us to integrate two different frameworks together.

Solution:(27th March)

- We converted Django to Node.js for backend.

01-04-2025:

Updated the Frontend of Chatbot in which I've added scrollbar, added a background to the input field.

02-04-2025:

1. Added functionality to send one prompt at a time
2. Done fine tuning of the chatbot

08-04-2025:

Added default prompts in chatbot.